
EEE 314

ANALOGUE ELECTRONICS

Department:	ELECTRICAL & ELECTRONIC ENGINEERING
Programme:	HND I
Year:	ONE
Semester:	THREE
Contact Hours:	5 hours/week

General Objectives

- 1.0 Know hybrid parameters and the application in the analysis of transistor amplifiers.
- 2.0 Understand the construction and principles of operation of semiconductor devices related to electrical power system.
- 3.0 Know the basic principles of operation and the applications of differential amplifiers.
- 4.0 Know the basic principles of operation and applications of operational amplifiers.

Assessment

Attendance = 5%

Assignments = 5%

Course Test = 10%

Practical = 20%

Examination = 60%

Course lecturer: **V.O. FACHE**

UNIT ONE

HYBRID PARAMETERS & APPLICATION IN THE ANALYSIS OF TRANSISTOR AMPLIFIERS

1.0 Introduction

A transistor, whether Bipolar Junction Transistor (BJT) or Field Effect Transistor (FET) is a three terminal semiconductor solid-state device used mainly as an amplifier to boost or raise the strength of a weak or small (amplitude) electrical signal. A transistor amplifier is used for signal amplification. Amplifiers make weak signal levels bigger or stronger and, in general, provide a function called gain. For example, a transistor (or audio) amplifier can be used to supply a strong signal to a loudspeaker.

A transistor amplifier consists of an active device a BJT or FET, d.c. source, a.c. source (input signal) and associated circuit components. To understand basic behaviour and characteristics of a small-signal transistor amplifier such as amplification, input and output resistance or impedance, voltage gain, current gain and power gain a two-port equivalent model of an amplifier is analysed using hybrid parameter (h-parameters).

1.1 Transistor Amplifier Circuit of BJT

A transistor amplifier circuit is used to raise the strength of a weak (small) signal and thus acts as an amplifier. Fig.1 shows the basic circuit of a transistor amplifier.

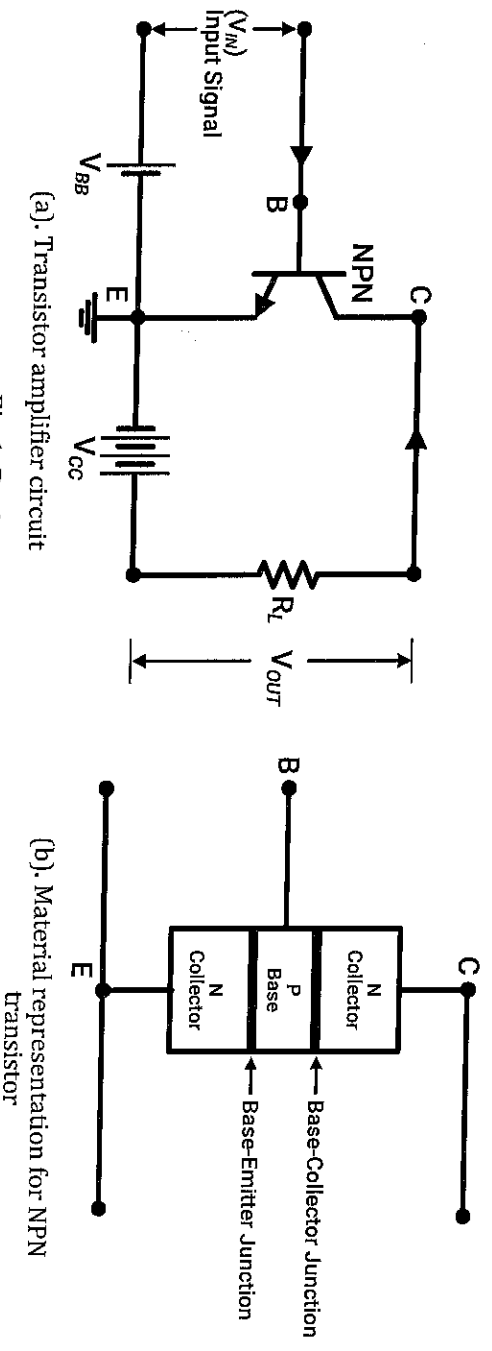


Fig.1. Basic circuit of transistor amplifier

The weak signal is applied between base-emitter (BE) junction and output is taken across the load R_L connected in the collector circuit. For a transistor to act as an amplifying device, the base-emitter (BE) junction should always remain forward biased and the base-collector (BC) junction should be reverse

biased. To do so, a d.c. voltage V_{EE} is applied to the input circuit in addition to the input signal V_{IN} as shown. This d.c. voltage is known as bias voltage and its magnitude is such that it always keeps the input circuit forward biased regardless of the polarity of the signal. The arrow on the emitter terminal of the transistor symbol shows the direction of forward current when the BE junction is forward biased.

As the input circuit has low resistance, therefore, a small change in signal voltage causes an appreciable change in emitter current. This causes almost the same change in collector current due to transistor action. The collector current flowing through a high load resistance R_L produces a large voltage across it. Thus, a weak signal applied in the input circuit becomes amplified in the collector circuit. It is in this way that a transistor acts as an amplifier.

Transistor amplifiers have three configurations or modes connecting the transistor as shown in Fig. 2.

- (1) Common emitter transistor configuration (CE transistor operation),
- (2) Common base transistor configuration (CB transistor operation) and,
- (3) Common collector transistor configuration (CC transistor operation).

In each mode, one of the three terminals of the transistor is grounded and the other two form input and output terminals. Thus, transistor amplifier is operated as a four terminal or a two-port network.

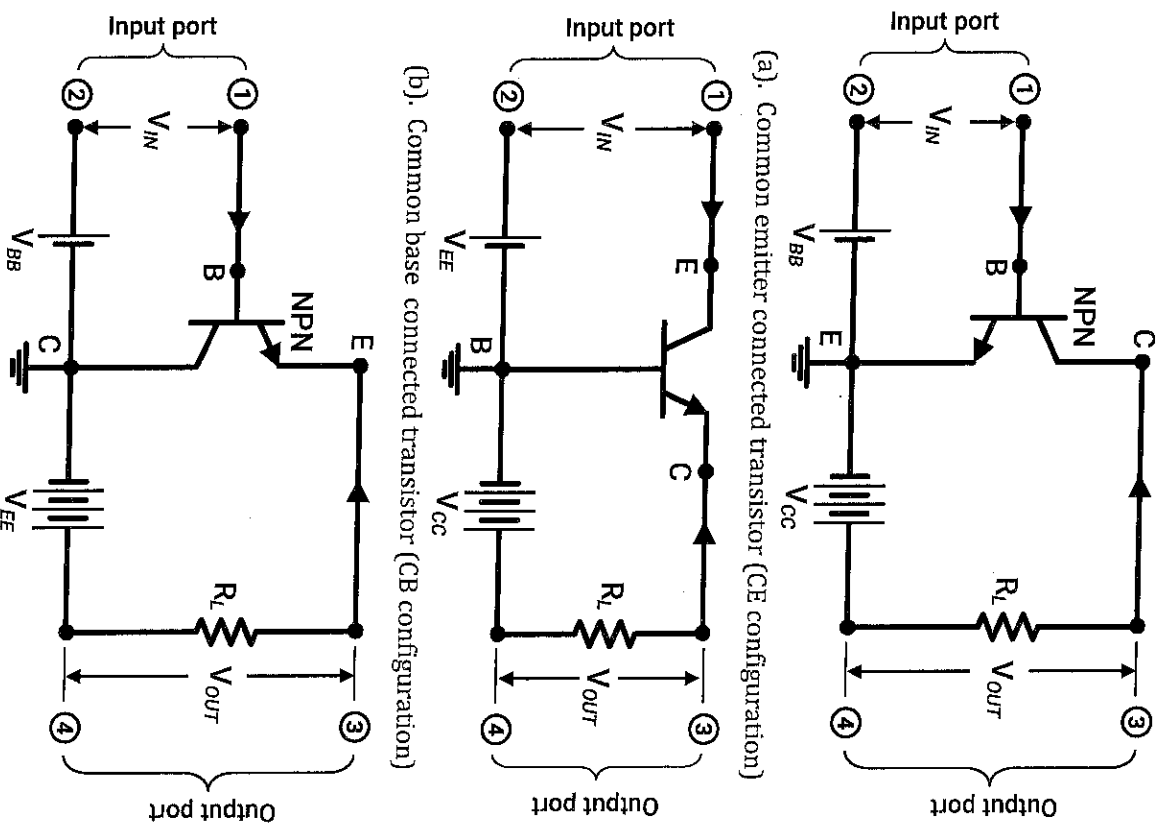
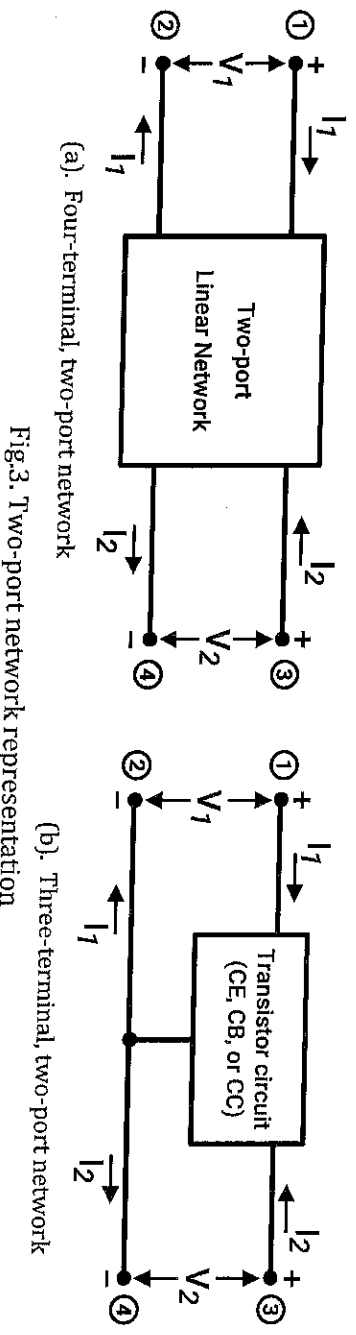


Fig. 2. Transistor amplifier configurations or modes

For instance, in common emitter configuration the emitter terminal is grounded and the input is applied between base and emitter forming the input port. Output is obtained between collector and emitter terminals, which form the output port. Thus, a transistor is considered as a two-port network as shown in Fig.2.

1.2 What is Hybrid Parameter?

Hybrid or h-parameter is a set of four mixed constants that can be used to represent the relationship between voltage and current in a two-port linear network or circuit. A linear network is one that the resistance, inductance, and capacitance do not change even when voltage across them is changed. They are called “hybrid” because they use mixed quantities (or units) to describe this relationship. The quantities are resistance or impedance (ohm), admittance (mho), voltage and current ratios or gains (no units). H-parameter is very useful in describing and analyzing two-port networks such as electronics circuits. **A two-port network is an electrical circuit or network with two separate ports for input and output.** Thus, a two-port network has four terminals; two terminals acting as input points and another two terminals acting as output points as shown in Fig. 3(a). Examples of four-terminal or two-port circuits are opamps, transformers and transistors. For a special case of three terminal devices such as transistor, the two-port network may appear as shown in Fig. 3(b). This simply shows that terminals 2 and 4 are in common.



H-parameter is used in the analysis and design of a transistor amplifier circuits to derive the gains and resistances. It is commonly used for the following reasons.

1. They are usually specified by component manufacturers.
2. They are easy to use in the analysis and design of transistor amplifier circuits.
3. They are much easier to measure experimentally than measure their impedance or admittance parameters.
4. They can easily be obtained from static characteristic curve in data sheets.

Because transistor amplifier can be considered as a two-port circuit means that its characteristics can easily be described using h-parameters. This method of analysis is used mainly for small-signal transistor amplifier circuits. Small-signal operation is that in which the input signal voltage and current are in the order of $\pm 10\%$ of the Q-point.

Consider the transistor (BJT) amplifier as two-port network represented by the block shown in Fig.3(b) with variables V_1 , I_1 , V_2 , I_2

From circuit theory, the voltages and currents can be related by the following sets of hybrid equations having a mixture of current and voltage variables on one side, as follows:

$$V_1 = h_{11} I_1 + h_{12} V_2 \dots\dots\dots(1)$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \dots\dots\dots(2)$$

In these equations, the **hs** are fixed constants called h-parameters. They are constant for a given transistor but change if the transistor is changed. Once these parameters are known, we can use equations (1) and (2) to find the voltages and currents in any amplifier circuit. To determine the parameters, both the short-circuit and open-circuit conditions are applied to Fig.3(b).

When the output port is short-circuited, V_2 becomes zero (0 V). Equations (1) and (2) are then used to get h_{11} and h_{21}

$$(1) \quad V_2 = 0, \text{ in Eqn.1.} \Rightarrow \text{Input resistance, } h_{11} = \frac{V_1}{I_1} \text{ (ohm)}$$

$$(2) \quad V_2 = 0, \text{ in Eqn.2.} \Rightarrow \text{Forward transfer current ratio, } h_{21} = \frac{I_2}{I_1} \text{ (no units)}$$

When the input port is open-circuited, I_1 becomes zero (0 A). Equations (1) and (2) are then used to get h_{12} and h_{22} .

$$(3) \quad I_1 = 0, \text{ in Eqn.1.} \Rightarrow \text{Reverse transfer voltage ratio, } h_{12} = \frac{V_1}{V_2} \text{ (no units)}$$

$$(4) \quad I_1 = 0, \text{ in Eqn.2.} \Rightarrow \text{Output admittance, } h_{22} = \frac{I_2}{V_2} \text{ (mho or siemens)}$$

It is evident from the equations that the h-parameters h_{11} , h_{12} , h_{21} , h_{22} , represent resistance or impedance, voltage gain, current gain and admittance respectively. This is why they are known as hybrid parameters.

When using h-parameters for applications, their numerical subscripts are replaced by the first letters defining them. Therefore,

$$\begin{aligned}
 h_{11} &= h_i = \frac{V_1}{I_1} & \Rightarrow & \text{Input resistance} \\
 h_{21} &= h_f = \frac{I_2}{I_1} & \Rightarrow & \text{Forward transfer current ratio} \\
 h_{12} &= h_r = \frac{V_1}{V_2} & \Rightarrow & \text{Reverse transfer voltage ratio} \\
 h_{22} &= h_o = \frac{I_2}{V_2} & \Rightarrow & \text{Output admittance}
 \end{aligned}$$

The hybrid equivalent circuit for equations (1) and (2) is shown in Fig. 4.

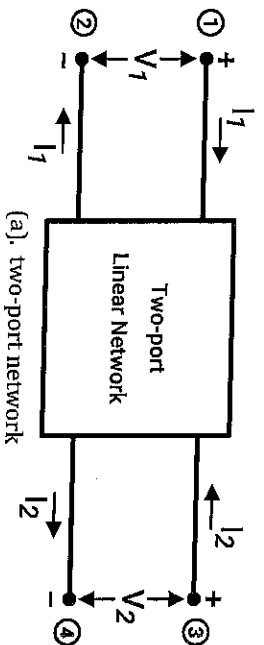


Fig.4. Hybrid equivalent circuit

